🧭 PROJECT OVERVIEW

🎯 Goal

Build a **browser-first passport & visa photo generator** that:

- Crops face correctly (rule-based)
- Removes background locally (no paid API dependency)
- Applies controlled lighting correction (no face alteration)
- Generates print-ready photos (future phase)
- Runs on **Next.js + Vercel free tier**

🧱 SYSTEM ARCHITECTURE

🟦 Frontend (Primary Processing Layer)

- Image upload
- Face detection
- Crop UI + adjustments
- Background removal
- Lighting correction
- Background selection
- Preview rendering

🟩 Backend (Minimal, Vercel API routes)

- Final image export (resize + format)
- Optional validation checks
- Print-sheet generation (later phase)

⚙️ CORE TECHNOLOGY STACK

🧠 Frontend Libraries

1. Face detection

- MediaPipe
  Used for:
- face bounding box
- eye position estimation
- auto crop initialization

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2. Crop UI

- react-easy-crop
  Used for:
- aspect ratio locked cropping
- drag/zoom adjustment
- preview-based editing

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3. Background removal

- @imgly/background-removal
  Used for:
- client-side segmentation
- human subject extraction

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4. Edge + image processing

- Canvas API (native browser)
- OpenCV.js (optional advanced layer)

Used for:

- feathering edges
- mask smoothing
- brightness/contrast adjustments
- histogram-based correction

🧾 Backend Libraries (Vercel API routes)

5. Final processing

- Sharp

Used for:

- final resize (passport standards)
- format conversion (JPEG)
- compression optimization

📐 SUPPORTED PHOTO FORMATS

🇮🇳 Indian Visa Photo

- Size: 2 × 2 inch
- Resolution: 300 DPI
- Pixels: 600 × 600
- Background: fixed white (#FFFFFF)

🇧🇩 Bangladesh Passport Photo

- Size: 1.5 × 1.9 inch
- Resolution: 300 DPI
- Pixels: 450 × 570
- Background: selectable (default blue)

🧠 CORE PROCESSING PIPELINE

STEP 1 — Upload

- User uploads 6–8MB image

- Immediate compression (client-side)

STEP 2 — Face Detection (MediaPipe)

- Detect:
  - face bounding box
  - eye position

STEP 3 — Smart Crop Engine

Rules:

- Lock aspect ratio (based on target format)
- Head occupies ~60–70% height
- Eyes aligned ~55–60% vertical position
- Center horizontally

STEP 4 — Manual Adjustment UI

Using react-easy-crop:

- drag image
- zoom in/out
- fine-tune alignment

STEP 5 — Background Removal

Using IMG.LY:

- run on cropped image only
- output transparent PNG

STEP 6 — Edge Refinement Layer

Apply:

- feather mask edges
- color decontamination
- optional OpenCV.js smoothing

STEP 7 — Background Replacement

India Visa:

- fixed white background

Bangladesh Passport:

- selectable palette:
  - blue (default)
  - light blue
  - white
  - off-white

STEP 8 — Auto Retouch (Lighting Correction)

Rules:

- brightness normalization
- gamma correction
- contrast adjustment
- white balance correction
- face-aware enhancement (MediaPipe region)

NO:

- skin smoothing
- face reshaping

STEP 9 — Quality Validation Engine

Checks:

- face size too small
- face off-center
- image too dark/bright
- blur detection

Output:

- warnings or auto-fix suggestions

STEP 10 — Final Export (Server-side)

Using Sharp:

- resize to exact pixel dimensions
- compress JPEG
- return downloadable image

🎨 UX FLOW DESIGN

Screen 1: Upload

- drag & drop / file select

Screen 2: Auto Crop Preview

- face detected overlay
- suggestion shown

Screen 3: Manual Adjust

- crop box (react-easy-crop)
- zoom slider

Screen 4: Background Selection

- color swatches (Bangladesh only)

Screen 5: Enhancement Toggle

- “Auto Enhance” ON/OFF

Screen 6: Final Preview

- before/after slider
- download button

🔥 KEY FEATURES (FINAL PRODUCT SCOPE)

✔ Core MVP

- Face detection
- Smart crop
- Background removal
- White/blue background
- Auto resize

✔ V1 Upgrade

- Manual crop adjustment
- Lighting correction
- Edge refinement

✔ V2 Premium

- Quality validation system
- Multi-background presets
- Before/after comparison UI

✔ V3 (future)

- Print-ready sheet generator
- Multi-photo layout (A4)
- Batch processing

⚠️ DESIGN CONSTRAINTS (IMPORTANT)

❌ Do NOT:

- beautify face
- reshape identity
- use heavy AI filters
- rely on external APIs for core pipeline

✅ DO:

- preserve identity
- correct lighting only
- keep processing deterministic
- prioritize speed + trust

🚀 ARCHITECTURE SUMMARY

UPLOAD
↓
MediaPipe (face detection)
↓
react-easy-crop (manual adjustment)
↓
IMG.LY (background removal)

↓
Edge refinement (Canvas/OpenCV.js)
↓
Background selection
↓
Auto lighting correction
↓
Sharp (final export on server)

🧠 PRODUCT POSITIONING (IMPORTANT)

Your app is NOT:

> ❌ photo editor

It IS:

> ✅ “passport-grade photo generator system”

That positioning is what makes it trustworthy and useful.